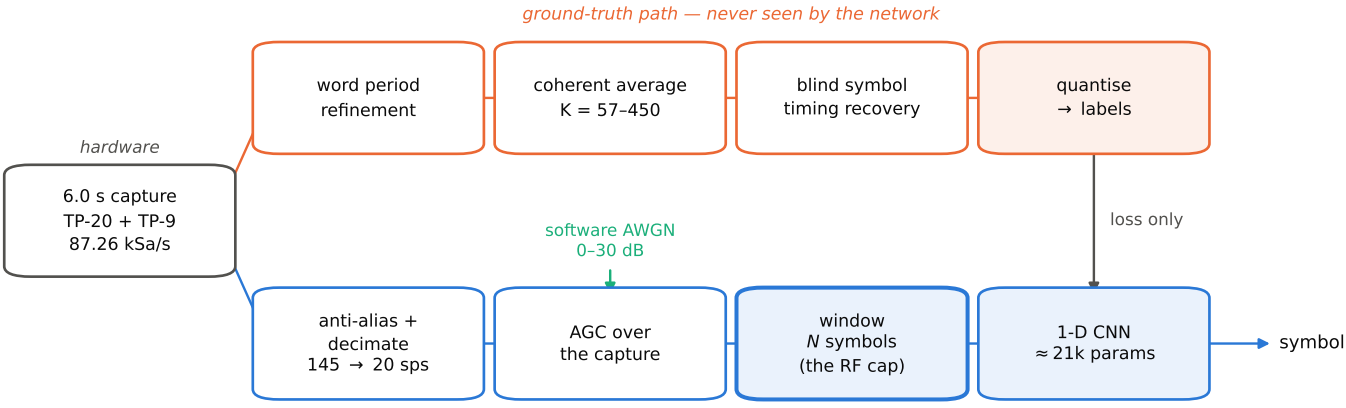
ground-truth path — never seen by the network
word period refinement
coherent average K = 57–450
blind symbol timing recovery
quantise → labels
hardware
6.0 s capture
TP-20 + TP-9
87.26 kSa/s
loss only
software AWGN 0–30 dB
anti-alias + decimate 145 → 20 sps
AGC over the capture
window N symbols (the RF cap)
1-D CNN ≈ 21k params
symbol
signal path — passband throughout; no carrier, pulse shape or constellation supplied